AI1 / AIML -- Revision

Supervised Learning

Content

- In this lecture we revise the content on Supervised Learning
- By solving the corresponding question from the Mock exam

- The Mock exam gives you a flavour of the style and difficulty of exam questions
- The question itself will be of course different at the exam

Question 2 Supervised Learning

(a) The following pseudo-code represents one iteration through the training set for gradient descent applied to univariate second order polynomial regression.

```
cost = 0;
w0 = 0;
w1 = 0;
w2 = 0;
for j in size(trainingSet)
        f = w0 + w1*x(j) + w2*x(j)^2
        cost = cost + (y(j) - f)^2
        w0 = w0 - a*(f - y(j))
        w1 = w1 - a*(f - y(j))*x(j)
        w2 = w2 - a*(f - y(j))*x(j)^2
endfor
```

Assume that the value of the learning rate, *a* is 1.

Give the numerical values of 'w0', 'w1', 'w2' and 'cost' at the end of this pseudo-code for the following training set: $\{(1,1),(2,5)\}$. Show all your working. **[10 marks]**

Solution for part a. -- thoughts

- To solve this question you need to know what is a training set -- so that you know what to do with the numbers given as “training set”. This was taught in the “Introduction to supervised learning” lecture.
- Then, following the steps of the pseudo-code gets the answer.
- You are expected to know what is regression, linear and polynomial regression – these were taught in the “Regression lecture” (even if in this particular question you don’t need to use that knowledge)

Solution for part a.

Iteration j=1:
w0=1, w1=1, w2=1, cost=1. Iteration j=2:
w0=-1, w1=-3, w2=-7, cost=5. These are the values at the end of the pseudo-code that the question was asking.

- Training data: {(1,1), (2,5)}. So that is, x(1)=1,y(1)=1, x(2)=2,y(2)=5.
- a=1, and before the loop we have: w0=0, w1=0, w2=0, cost=0.
- Iteration j=1:

```
f = w0 + w1*x(j) + w2*x(j)^2   = 0 + w1*x(1)+w2*x(1)^2 = 0+0*1+0*1^2 = 0.
cost = cost + (y(1)-f)^2       = 0 + (y(1)-f)^2        = 0+(1-0)^2   = 1.
w0 = w0 - a*(f-y(j))           = 0 - 1*(f-y(1))        = 0-(0-1)     = 1.
w1 = w1 - a*(f-y(j))*x(j)      = 0 - 1*(f-y(1))*x(1)   = 0 - (0-1)*1 = 1.
w2 = w2 - a*(f-y(j))*x(j)^2    = 0 - 1*(f-y(1))*x(1)^2 = 0 - (0-1)*1^2 = 1.
// So, after 1st iteration we have: w0=1, w1=1, w2=1, cost=1.
```

- Iteration j=2:

Solution for part a.

Iteration j=1:
w0=1, w1=1, w2=1, cost=1. Iteration j=2:
w0=-1, w1=-3, w2=-7, cost=5. These are the values at the end of the pseudo-code that the question was asking.

- Training data: {(1,1), (2,5)}. So that is, x(1)=1,y(1)=1, x(2)=2,y(2)=5.
- a=1,

  // So, after 1st iteration we have: w0=1, w1=1, w2=1, cost=1.

- Iteration j=2:

  f = w0 + w1*x(j) + w2*x(j)^2 = 1 + w1*x(2)+w2*x(2)^2 = 1+1*2+1*2^2 = 7.
  cost = cost + (y(1)-f)^2 = 1 + (y(2)-f)^2 = 1+(5-7)^2 = 5.
  w0 = w0 - a*(f-y(j)) = 1 - 1*(f-y(2)) = 1-(7-5) = -1.
  w1 = w1 - a*(f-y(j))*x(j) = 1 - 1*(f-y(2))*x(2) = 1-(7-5)*2 = -3.
  w2 = w2 - a*(f-y(j))*x(j)^2 = 1 - 1*(f-y(2))*x(2)^2 = 1-(7-5)*2^2 = -7.
  // So, after 2nd iteration we have: w0=-1, w1=-3, w2=-7, cost=5.
  And we are done!

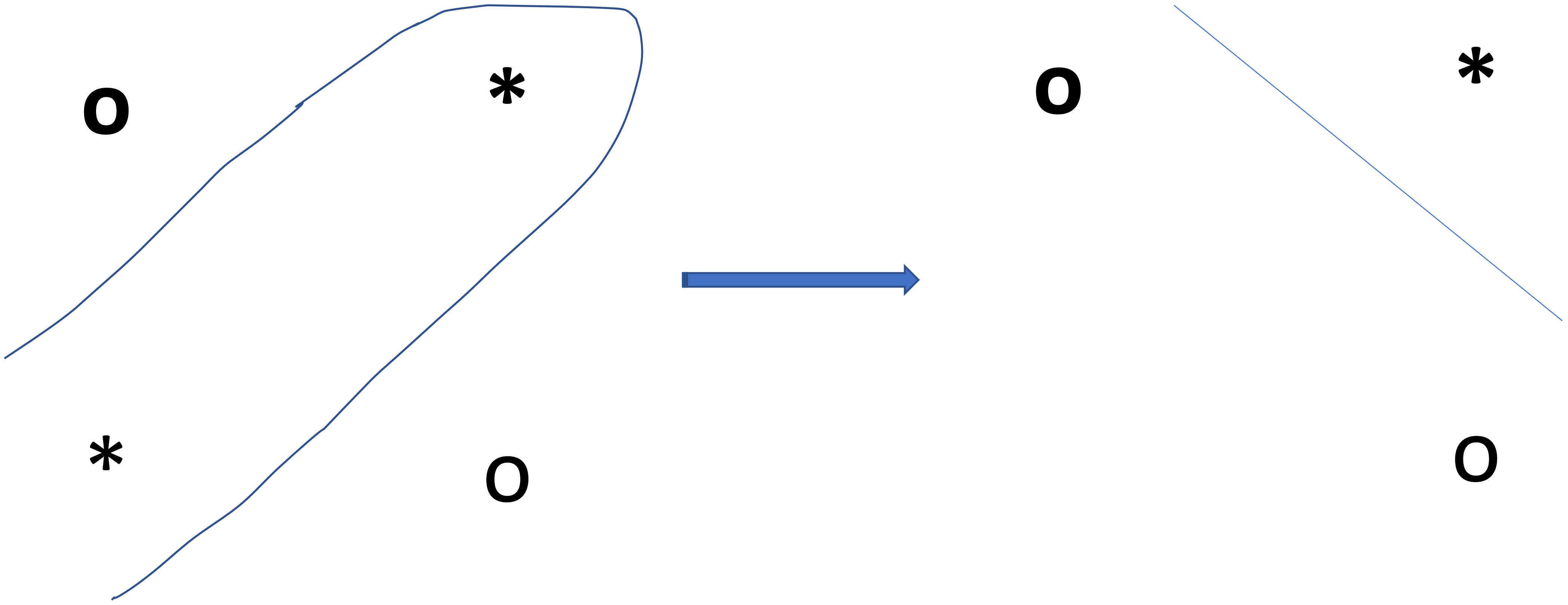
(b) Consider a multivariate data set with 2 classes that are not linearly separable. Is it true that the classes will still be not linearly separable

- (i) if you remove one point from this data?
- (ii) if you remove one feature from this data?

In both cases, justify your answers in the following way: if your answer is yes, then explain why; if your answer is no then give a counter-example. **[10 marks]**

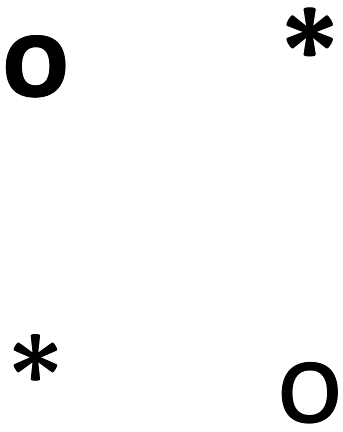
Solution for part b.(i)

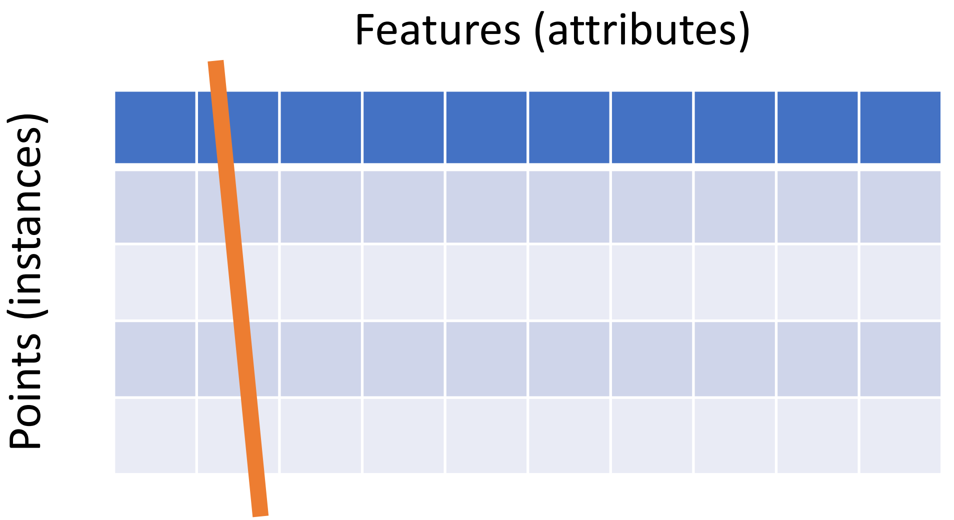
(i) No: If you remove a point it can become linearly separable. For example, the following 2D labelled data set ((-1,-1), 0), ((1,1), 0), ((-1,1),1), ((1,-1),1) is not linearly separable, but after removing (any) one point from it it becomes linearly separable. (Draw the data for yourself to see.)



Solution for part b.(ii)

(ii) Yes, it remains not linearly separable. That is, a weight vector with which a linear classifier would perfectly label this data does not exist. The reason is a follows: Removing a feature is equivalent to setting the component of the weight vector that multiplies that feature to 0. If this would make the classes linearly separable then this modified weight vector would perfectly label the original data set, which is a contradiction.





$$\boldsymbol{w}^T\boldsymbol{x} = w_0 \cdot 1 + w_1 x_1 + \cancel{w_2 x_2} + \cdots * w_d x_d$$

Imagine deleting any one of the features, say x_2

Revise

- Anything taught can be asked.
- Anything that can be deduced, by thinking, from what has been taught can be asked.
- Nothing outside taught material will be asked.
- Optional material will not be asked – but it helps you gain deeper understanding, and general thinking skills.
- Revise! Pay attention to all worked examples.
- No memorising required, but a good understanding is required.

Before you go…

- As you can see, you don’t need to know calculus/ differentiation to answer this year’s exam-style questions (but if you do, it gives you a more complete and deeper understanding).

- Please give us some feedback here:

https://canvas.bham.ac.uk/courses/65652/quizzes/170557

Good luck!